Electrochemical C-N Bond Formation Within Boron Imidazolate Cages Featuring Single Copper Sites

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ABSTRACT: Electrocatalysis expands the ability to generate industrially-relevant chemicals locally, on-demand, using intermittent renewable energy, thereby improving grid resiliency, and reducing supply logistics. Herein, we report the feasibility of using molecular copper boron-imidazolate cages, BIF-29(Cu), to enable coupling between the electroreduction reaction of CO_2 (CO_2RR) with NO_3^- reduction (NO_3RR) to produce urea with high selectivity of 68.5% and activity of 424 μA cm^{-2}. Remarkably, BIF-29(Cu) is among the most selective systems for this multi-step C-N coupling to-date, despite possessing isolated single-metal sites. The mechanism for C-N bond formation was probed with a combination of electrochemical analysis, in-situ spectroscopies, and atomic-scale simulations. We found that NO_3RR and CO_2RR occur in tandem at separate copper sites with the most favorable C-N coupling pathway following the condensation between $^*CO$ and NH_2OH to produce urea. This work highlights the utility of supramolecular metal-organic cages with atomically discreet active sites to enable highly efficient coupling reactions.

Introduction

The electrochemical utilization of environmental pollutants offers a promising means for synthesizing valuable chemicals while also counteracting industrial emissions. Specifically, the electrochemical reduction of carbon dioxide (CO_2) has increased in technical design and efficiency in recent years to synthesize high-value products including carbon monoxide (CO), formate ($HCOO^-$), methane (CH_4), ethylene (C_2H_4), methanol (CH_3OH), and ethanol (C_2H_5OH).[1,2] There has been recent interest in furthering the synthetic complexity of CO_2RR product distributions with the introduction of heteroatom bond formations, with a specific focus on C-N bond forming reactions.[3–5] A particular C-N containing molecule of interest is urea, an industrial chemical that is produced on the scale of 200 million tons annually.[6] Urea is primarily used in the agricultural industry as a nitrogen-dense fertilizer that readily decomposes to ammonia (NH_3) and CO_2. The primary issues with the current industrial urea production are its association to the Haber-Bosch process which possesses high energy requirements (>2% global energy supply) and is associated as one of the largest industrial contributors to CO_2 emissions.[7] Furthermore, the high degree of global nitrogen fixation has resulted in the concentration of harmful environmental pollutants within aquatic ecosystems (NO_x).[8] With this backdrop, electrochemical urea production appears to be a viable option to utilize key environmental pollutants (NO_3^-/NO_2^- and CO_2) to synthesize a necessary industrial chemical.

Recently, many electrochemical platforms capable of enabling C-N bond formation have accomplished impressive selectivity and activity.[4] As a result, researchers have leveraged several material design factors to mediate electrochemical heteroatom bond formation over heterogenous active sites, including electronic effects[9], vacancy/defect sites[10], and catalyst supports[11]. However, mechanistic insights into these heterogeneous systems are often difficult to obtain, limiting further synthetic adaptation. We have therefore focused our attention on systems that engender molecular tunability and well-defined active sites to probe the reaction mechanisms. To this end, we chose to explore metal-organic frameworks (MOFs) as a modular platform with which to isolate molecular ac tive sites.[12]

Boron-imidazolate frameworks (BIFs), are a subset of MOFs that are composed of metal-imidazole-boron linkages. These materials are credited for having versatile structural topologies[13] while also maintaining high chemical[14] and thermal stability.[15] Recently, our group has shown that BIF-29(Cu), a copper-containing BIF, and its thermolyzed derivative produce methane from CO_2RR with high selectivity (32 and 55%, respectively).[15] Extrapolating from these results, we infer that the propensity to form CH_4 in high yield suggests a relatively strong $^*CO$ intermediate that could be leveraged for electrochemical C-N bond formation.[16] Despite containing isolated metal sites, BIF-29(Cu) can catalyze the formation of C-N bonds from CO_2 and NO_3^-, exhibiting a high Faradaic efficiency for urea production of 68.5 ± 0.57 % with a partial current density of 424 ± 16 μA cm^{-2}. Surface enhanced infrared absorbance spectroscopy (SEIRAS) aided in our investigation of the reaction mechanism and showed the addition of nitrate to the system intercepts the CO_2RR pathway by forming an $^*NH_2CO$-containing species. Surveying the substrate scope of this reaction suggests that this intermediate is formed from a catalytic cascade reaction where NO_3^- is first reduced to hydroxylamine (NH_2OH), which can then react with the adsorbed $^*CO$ intermediate to form the first C-N bond. A repeat of this step, in concert with two proton/electron transfer events, yields the final urea product. We sought to probe the active site further through *in situ* X-ray absorbance spectroscopy (XAS), where we observed dramatic changes in coordination environment under potential bias and electrolyte conditions. Interestingly, we observe that the structural perturbations of the active site were minimized in the presence of both CO_2 and NO_3^-, as both substrates work in tandem to facilitate turnover of the active site. Density functional theory (DFT) calculations are further carried out to explore the mechanistic steps for C-N coupling to form urea at these molecular sites.[17] Consistent with experimental observations, DFT calculations support isolated NO_3RR and CO_2RR at different active sites with the initial C-N union occurring between $^*CO$ and NH_2OH at a single Cu site.[18] This works establishes the utilization of supramolecular metal-organic cages for the confinement and condensation of reduction intermediates to enable highly efficient asymmetric coupling reactions.

Results and Discussion:

Synthesis and characterization of the BIF-29(Cu) material. The boron-imidazolate framework utilized in this work, BIF-29(Cu), is composed of discrete metal-organic cages following an M_6L_8 stoichiometry (M = Cu, L = $HB(Im)_3$, where Im is imidazole). The isolated Cu active sites within these cages exist in a square pyramidal geometry with a weakly coordinated axial water molecule. The powdered BIF-29(Cu) material was synthesized solvothermally between a $Cu(NO_3)_2$ salt and the tridentate $Na[HB(Im)_3)]$ ligand. The identity of the as-synthesized purple solid was confirmed by powder X-ray diffraction (PXRD, **Figure 1A**, purple trace). Infrared absorption spectroscopy (FTIR), **Figure 1D**, red trace) showed characteristic B-H (2,426 cm^{-1}), imidazole C-H (3,132 cm^{-1}), and axial water O-H (3,435 cm^{-1}) bands. Previous reports have illustrated the feasibility of dissolving the resulting framework material into colloidally stable solutions by disrupting the van der Waals interactions between cages.[15,19] This unique feature of the framework material lends itself to versatile processability and incorporation into electrochemical systems. The colloidal stability of a 1 mg mL^{-1} MeOH solution of BIF-29(Cu) was confirmed by dynamic light scattering (DLS) measurements, which showed an average particle size of 300 nm (**Figure S1**). SEM imaging showed that the bulk crystal material was primarily composed of ~5.0 μm octahedral crystallites (**Figure 1C,** top), while the morphology of the drop casted BIF-29(Cu) sample proved to be substrate dependent. On a glass slide, the resulting material appeared largely amorphous (**Figure 1C**, lower left). When the BIF-29(Cu) solution was deposited onto a carbon paper substrate, submicron crystallites were observed (**Figure 1C**, lower middle). Interestingly, when the BIF-29(Cu) solution was mixed in a 1:1 mass ratio of <7 nm multi-walled carbon nanotubes (MWCNT), the BIF cages recrystallize to their original octahedron-based geometry (**Figure 1C**, lower right). Extended SEM images can be found in **Figure S2**. PXRD (**Figure 1A**, blue trace) and FTIR (**Figure 1D**, black trace) of the drop-cast BIF-29(Cu) material confirms that the preferred crystallographic orientation and composition are retained relative to the bulk material. Owing to the enhanced electrochemical integration of BIF-29(Cu) on the conductive MWCNT and the well-controlled particle morphology and size, we employed BIF-29(Cu)/MWCNT as an electrocatalyst for urea production.

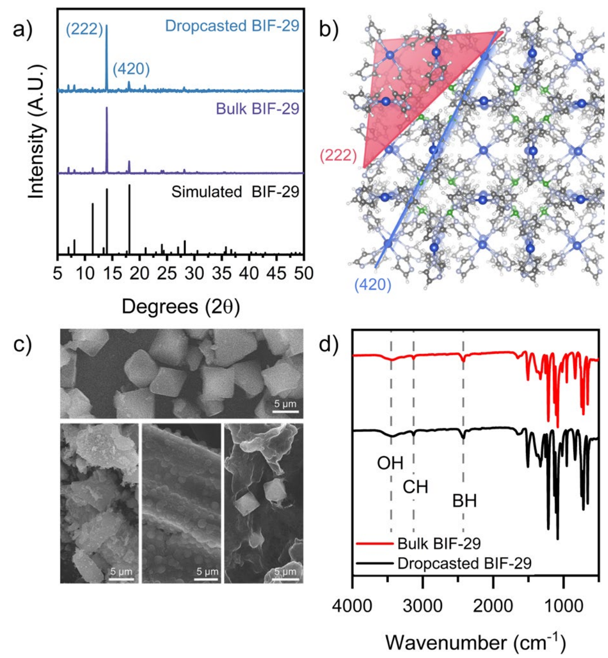

Figure 1. (a) Powder X-ray Diffraction of the bulk and drop-casted BIF-29(Cu) materials compared to the simulated PXRD of the BIF crystal. (b) BIF-29(Cu) unit cell scheme showing the atomic arrangements along the two principal diffraction planes. (c) Scanning Electron Microscopy of the bulk BIF-29(Cu) material (top), the drop-casted BIF-29 on glass (lower left), the drop casted BIF-29(Cu) on carbon paper (lower middle), and the recrystallized BIF-29(Cu) in the presence of multi-walled carbon nanotubes (lower right). (d) FTIR characterization of the bulk and drop-casted BIFs, plotted in transmittance.

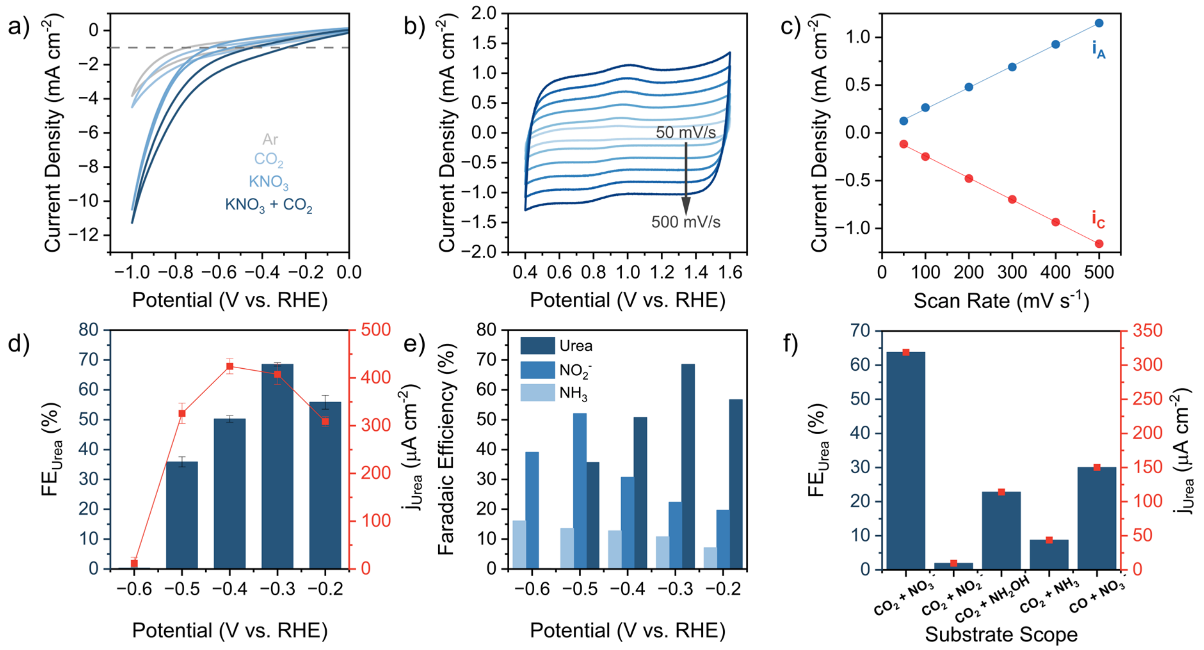

Figure 2. (a) Cyclic voltammetry response of BIF-29(Cu) supported on carbon paper in 0.1 M $KHCO_3$ and 0.1 KNO_3 under either CO_2 or Ar using a scan rate of 0.1 mV s^{-1}. (b) Scan rate dependent CVs of BIF-29(Cu) on carbon paper in 0.1 M $KHCO_3$. (c) Plotted peak current densities for the anodic (i_A) and cathodic (i_C) copper redox features as a function of the scan rate. (d) Electrochemical selectivity and activity of BIF-29(Cu)/MWCNT/CP towards urea production across several potentials in 0.1 M $KHCO_3$ + 0.1 M KNO_3 and (e) the corresponding liquid product distributions. (f) Electrochemical urea production metrics with different carbon and nitrogenous precursors after galvanostatic electrolysis at -0.5 mA cm^{-2}.

Electrochemical studies for urea production using a BIF-29(Cu)/MWCNT composite. Our investigation into the electrochemical C-N bond formation activity of this metal-organic cage material began with observing the voltametric response to the presence of different substrates (**Figure 2A**). Activity increases within the range of 0.0 to -1.0 V vs. the reversible hydrogen electrode (RHE) upon the addition of either CO_2 or NO_3^-, suggesting competitive reaction mechanisms against the parasitic hydrogen evolution reaction (HER). When both CO_2 and NO_3^- are present, there is a further shift in the catalytic onset to -0.3 V vs. RHE, signifying a cooperative C-N coupling mechanism at lower potentials. All liquid-phase products were quantified through colorimetric methods (**Figures S3-6**). Electrolysis under continuous flow CO_2 showed that the BIF-29 cage was capable of synthesizing urea with a high Faradaic efficiency of 22% and partial current density of 192 μA cm^{-2} at -0.5 and -0.6 V vs. RHE, respectively (**Figure S8**). Utilizing scan-rate dependent cyclic voltametric measurements within the $Cu^{II/I}$ redox window, we found that the BIF-29(Cu) material deposited onto carbon paper resulted in low electrochemically active copper sites, EA_{Cu} = 0.26 nmol cm^{-2} (**Figure 2B, C**). To combat the low density of active sites, we supplemented our electrode composite with multiwalled carbon nanotubes (MWCNT). With gravimetric ratios of 10:0.5, 10:1, and 10:2 (CNT:BIF-29), we observe EA_{Cu} values of 6, 8, and 10 nmol cm^{-2}, respectively (**Figure S9**). Constant potential electrolysis (CPE) for each ratio yielded higher selectivity and activity metrics. The highest performance was achieved using the 10:2 composite ratio material with a urea FE = 54.8% and j_{urea} = 329 μA cm^{-2} at -0.4 vs. RHE (**Figures S10-11**).

Having optimized the electrode composite, we applied these conditions across a potential range and found high selectivity and activity between -0.2 to -0.5 V vs. RHE with a steep drop off at -0.6 V vs. RHE (**Figure 2D**). Optimal urea selectivity was achieved at a potential of -0.3 V vs. RHE with a Faradaic efficiency of 68.5 ± 0.57 % and current density of 407 ± 21 μA cm^{-2}. At -0.4 V vs. RHE, we see a slight decrease in efficiency but achieve our highest specific current density for urea of 424 ± 16 μA cm^{-2}, an activity that is competitive with other urea forming electrocatalysts (**Figure 2D**). Product distributions across this potential range show nitrite as the second major product after urea between -0.2 and -0.4 V vs. RHE with an inversion in selectivity at -0.5 V vs. RHE (**Figure 2E**). Electrolysis with solely the nanotube support showed negligible activity for NO_3RR (**Figure S12A**).

To evaluate the active intermediates in our C-N coupling step, we performed electrolysis with different isolatable intermediates in conventional CO_2RR and NO_3RR mechanisms (**Figure 2F**). With respect to the C-based intermediates, both CO_2 and CO prove to be effective at producing urea, although CO exhibits only half the selectivity and activity for urea production likely due to its lower water solubility.[20] For the N-based intermediates, only hydroxylamine shows a significant capacity for urea production, albeit at a reduced selectivity and activity likely because hydroxylamine is prone to hydrolysis in water. Notably, hydroxylamine was detected in modest yields under catalytic conditions in the presence CO_2 and NO_3^- (**Figure S8B**) under un-optimized electrode conditions, which suggests it is a viable intermediate for C-N coupling. Ammonia was observed to have relatively low selectivity for urea, although it has been used previously as a nucleophile in C-N coupling over copper;[5]

the reaction between ammonia with carbon-based intermediates likely follow a different pathway than the N-oxygenate substrates evaluated herein. Pulsed-potential experiments show a decrease in urea selectivity from 68.5% to 46.1%. We attribute the decrease in selectivity upon pulsing the potential anodically to the diffusion of partially reduced intermediates away from the electrode surface (**Figure S12B**). Coincidentally, we notice that, with pulsing, we shift selectivity towards nitrite, suggesting that nitrite may play an important role in the formation of hydroxylamine which is the key nitrogenous species responsible for urea production. Upon optimization, the BIF-129(Cu) material achieved a turnover number (TON) for urea up to approximately 200 at -0.3 V vs. RHE (**Figure S13**).

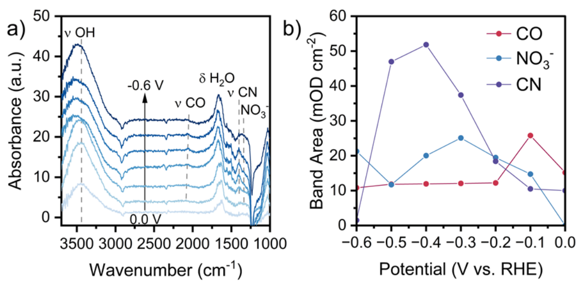

Figure 3. (a) Potential dependent SEIRAS spectra of BIF-29(Cu) drop cast onto a Au coated Si wafer immersed in a solution of 0.1M $KHCO_3$ and 0.1M KNO_3 (sat. CO_2, pH = 6.8). Each spectrum was collected following background subtraction at 0.4 V vs. RHE. (b) Integrated band areas for the CO, NO_3^-, and CN stretches.

Probing electrochemically generated species via *in-situ* SEIRAS. To validate our electrochemical experiments and develop a better understanding of the reaction mechanism, we probed the interfacial species being generated at the electrode-electrolyte interface using surface-enhanced infrared absorbance spectroscopy (SEIRAS). The assembled SEIRAS setup can be seen in the supplemental information (**Figure S14**). The SEIRAS electrode was prepared by thermal evaporation of a 50 nm Au thin film onto a silicon ATR wafer followed by drop-casting the BIF-29(Cu) methanol solution directly onto the Au/Si wafer until a 0.2 mg cm^{-2} loading was achieved. The obtained BIF-29(Cu)/Au/Si film was then dried in a vacuum oven for 3 hours at 50 ^{o}C to ensure complete activation of the MOF film. Electrochemical CO_2RR in the absence of any nitrogenous substrate showed the formation of an adsorbed *CO species that is stable until reaching -0.8 V (**Figure S15**). This CO band is believed to arise from a Cu-CO adsorbate since Au does not concentrate *CO under a negative bias.[21]Upon the addition of nitrate, electrochemical C-N bond formation can be visualized with an increasing IR band between 1,410 and 1,415 cm^{-1}, which is consistent with other reports (**Figure 3A,B**).[4] To validate the chemical identity of this peak, we collected a library of solution phase IR spectra of different isolatable species anticipated to form during electrolysis (**Figure S19, Table S1**)). This data supports our initial assignment that the C-N species we observe arises from an intermediate resembling carbamate [$COOH(NH_2)$], the same C-N coupling intermediate in industrial urea production. This is further substantiated by the atomic-scale simulations of our anticipated C-N intermediate with two steps for the urea formation by one *CO and two NH_2OH species (*vide infra*). To verify that these species observed by IR arise from BIF-29(Cu) rather than the Au sublayer, we repeated these experiments on a freshly prepared Au/Si wafer and found that although the C-N band can still be observed, its intensity is approximately forty times lower than with the BIF-29(Cu)/Au/Si electrode composite (**Figure S16**). Furthermore, we tested the stability of the molecular film by performing cyclic voltammetry between highly cathodic potentials and observed a retention of the C-N band at intermediate potentials for all three scans (**Figure S17**).

Observing the local structure of the Cu sites under a potential bias using x-ray spectroscopy. In order to evaluate the electronic structure and local coordination environment of the CuN_4 sites under electrolytic conditions, we performed *in situ* X-ray absorption spectroscopy (XAS). XAS was conducted under different experimental conditions to decouple the competing reactions of HER, NO_3RR, CO_2RR, and C-N coupling. In the X-ray absorbance near edge structure (XANES), we observe different trends in the white line intensities and the rising edge under the different experimental conditions at OCP (**Figure S20**). Across the NO_3RR, CO_2RR, and the C-N coupling conditions we see an increase in the white line intensity indicating a decrease in electron density around the Cu site, likely arising from the formation of electron withdrawing adsorbates. Under CO_2RR condition, we also observe a pre-edge feature at open circuit suggesting new coordination geometry, consistent with the DFT optimized structure for the CO-CuN_4 complex, where copper is pulled out from its planar N_4 coordination environment (**Figure S32**). With increasing cathodic bias, we observe a decrease in white line intensity, as the electron density of the Cu site increases, coupled with the growth of different rising edge features across all substrate conditions, suggesting a preactivation of the Cu(II) active sites to Cu(I) followed by distortions in the local geometry of the active site (**Figure S21-24**).

The local coordination environment of Cu was further evaluated through the extended X-ray absorbance fine structure (EXAFS). The EXAFS data further confirm a CuN_4 coordination environment with the anticipated Cu-N scattering at 1.5 Å (phase corrected) and coordination numbers averaging around 4.4 ± 0.35 at open circuit potential for all experimental conditions. .[22] All k^2-space fittings are expressed in **Figures S25-28**. In modelling the EXAFS, we found that in the absence of CO_2 and NO_3^- substrates, a decrease in Cu-N scattering was accompanied by a simultaneous increase in Cu-Cu coordination with increasing cathodic bias (**Figure 4A, Table S2)**, which suggests the CuN_4 sites within the M_6L_8 cages dissociates in the absence of substrate. We attribute this phenomenon to a build-up of charge at the copper sites that cannot effectively dissipate, resulting in Cu abstraction from the N_4 coordination environment enabling the nucleation of metallic Cu clusters.[23] Interestingly, we find that this process is semi-reversible under all conditions; the Cu-Cu coordination decreases as we step the applied potential back to OCP with a corresponding rise in Cu-N coordination. With the addition of nitrate under a N_2 atmosphere, we observe a slower progression to Cu-Cu coordination with increasing cathodic bias (**Figure 4B, Table S3**). With CO_2 in the absence of nitrate, we observe no Cu-Cu bond formation; however, there is significant

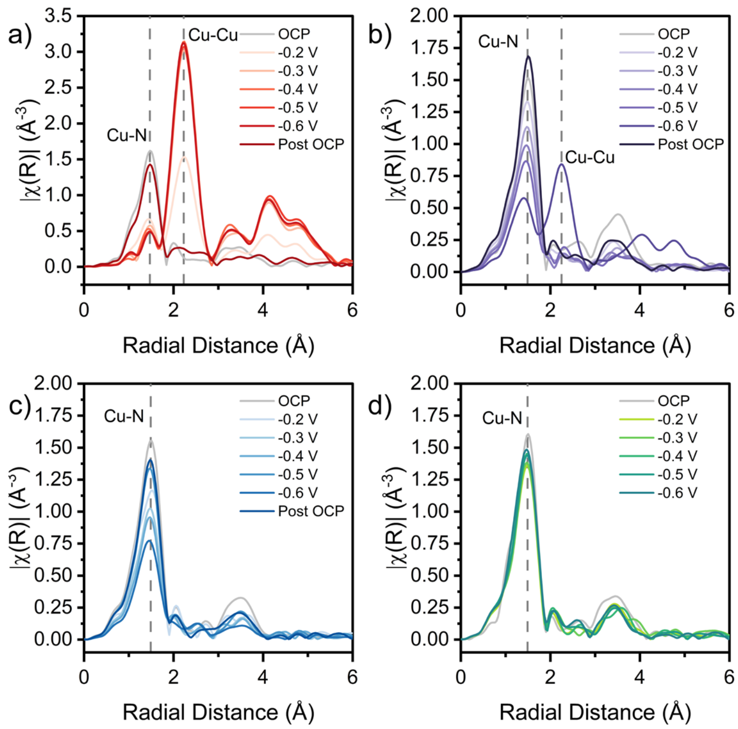

Figure 4. Extended X-ray Absorption Fine Structure (EXAFS) of BIF-29(Cu) at different potentials in (a) 0.1M $KHCO_3$ (N_2 purged), (b) 0.1M $KHCO_3$ and 0.1M KNO_3 (N_2 purged), (c) 0.1M $KHCO_3$ (CO_2 purged), and (d) 0.1M $KHCO_3$ and 0.1M KNO_3 (CO_2 purged).

attenuation of the Cu-N coordination number with increasing cathodic potential (**Figure 4C, Table S4**). We propose that this decrease arises from the formation of a strong *CO adsorbate on Cu that cannot effectively turnover due to the low applied potentials (**Figure S58**).[24] Under experimentally relevant C-N coupling conditions, the BIF-29(Cu) active site is relatively stable at all potentials: we observe no Cu-Cu formation, and the Cu-N coordination number does not significantly change, signaling effective turnover of the active site (**Figure 4D, Table S5**). We do note a new rising edge feature consistent with the formation of a Cu(I) species.[25] In summation, the *in situ* XAS results show that the molecular Cu site remains intact during the catalytic co-reduction of CO_2 and NO_3^- to urea.

Computational evaluation of the cascade mechanism for urea production. After confirming stability under catalytically relevant conditions, we surmised that electrochemical C-N bond formation is likely to follow a cascade-type mechanism at individual copper sites isolated within the molecular cage structure (**Figure S29**). Therefore, at least one of our reduction intermediates must desorb from one site and chemically coalesce at another active site before desorbing as urea (**Figure 5A**). To this end, we sought to evaluate the possible elementary steps for CO_2RR, NO_3RR, and C-N coupling to form urea (**Figure S30)**. DFT optimized adsorption configurations of possible intermediates and their corresponding adsorption energies were calculated and are shown in **Figures S31-S54** and **Table S6**. The Gibbs free energy diagrams for each possible NO_3RR pathway are presented in **Figures S55-S61**. Previous literature for CO_2RR over Cu-based single atom catalysts have reported that the rate determining step is the first proton coupled electron transfer (PCET) with CO_2 to form *COOH at the single Cu site.[26,27] The second proton/electron would then transfer to *COOH to form *CO and water, a stable reduction intermediate that was observed experimentally (**Figure 3B, S15**). We found that potential rate determining step of CO_2RR-to-*COOH is nearly 0 eV under applied potential of -0.20 V vs. RHE and pH of 6.8. Furthermore, the reaction energetics of CO_2RR-to-*CO is -1.80 eV and the adsorption of the *CO over the single Cu site is chemisorption with a Gibbs adsorption energy of -1.20 eV (**Figure S62**). The results agree well with Yang et al. that CO_2RR to CO over Cu-based single atom catalyst is exothermic with *COOH as the intermediate.[28]As hydroxylamine (NH_2OH) has been experimentally identified as a viable precursor to urea, various reaction pathways for NO_3RR-to-NH_2OH were investigated. Based on our calculated free energy diagrams, the most favorable reaction mechanism followed the first PCET step to the adsorbed *NO_3 to form *NO_3H. The second PCET step with *NO_3H would result in *NO_2. Since the adsorption configurations of the following NO_2RR-related intermediates at the single Cu site may be either through the Cu-O or the Cu-N bond, we performed both possible adsorption configurations for the NO_2RR-related intermediates and found that these intermediates preferentially adsorb over the single Cu site through the Cu-N bond. The Gibbs free energy diagrams of NO_2RR-to-NH_2OH show that the most favorable pathway is through the intermediates *NO_2H and *NO via two exothermic proton transfer steps. Finally, the key intermediate *NO was

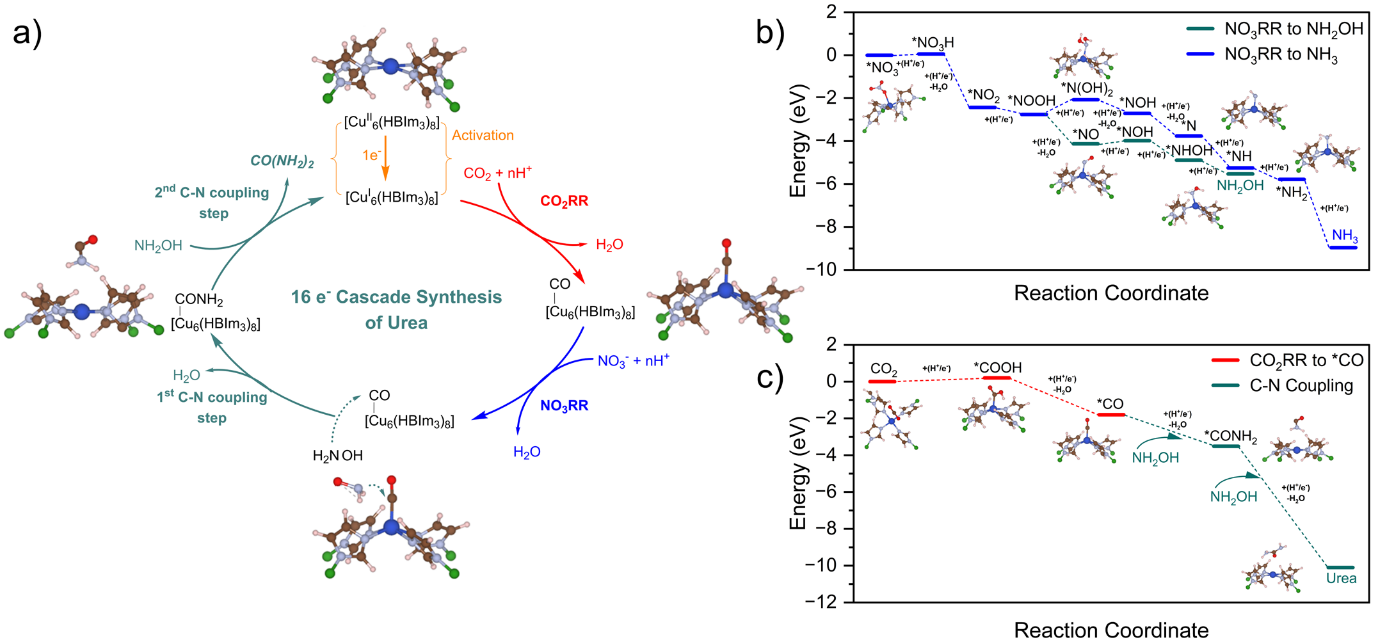

Figure 5. (A) The proposed mechanism for urea production from CO_2 and NO_3^- within the BIF-29(Cu) cage and (B) Gibbs free energy diagrams for NO_3RR to hydroxylamine or ammonia (C) Gibbs free energy diagram of CO_2RR-to-CO and hydroxylamine over the single Cu atom site within the structure of BIF-29(Cu) with applied potential of -0.20 V vs. RHE and pH set as 6.8.

further hydrogenated to form $^*NH_2OH$ via another three exothermic PCET steps. Furthermore, $^*NH_2OH$ adsorption at the single Cu site was via physisorption with a Gibbs adsorption energy of 0.05 eV. Therefore, $^*NH_2OH$ could easily desorb from the Cu active site once formed. This result agrees well with the experimental observations that the NO_3RR and CO_2RR are first reduced in tandem at separate Cu sites to form the potential key intermediates $^*CO$ and NH_2OH. With these two key intermediates, $^*CO$ would be retained at the single Cu site, while the weakly bonded NH_2OH would desorb from the Cu site and coalesce with $^*CO$ to form the final product urea $CO(NH_2)_2$. The different reaction pathways to generate NH_2OH and $^*CO$ following NO_3RR and CO_2RR under the applied potential of -0.20 V vs. RHE and pH set as 6.8 are summarized in **Figures 5B and 5C**.

According to the Gibbs free energy diagrams in **Figure 5B** and **5C**, the Gibbs reaction energy for $CO(NH_2)_2$ production with co-formation of $2H_2O$ via C-N coupling between $^*CO$ and $2NH_2OH$ is -4.53 eV (-2.26 eV per C-N bond coupling with co-formation of one H_2O). The Gibbs reaction energy of $NH_2OH+{*}+H^++e^- \leftrightarrow {}^*NH_2+H_2O$ is around -1.65 eV, which is less thermodynamically favorable than that of urea formation. Furthermore, we calculated the activation barrier of the C-N coupling process, i.e., $^*CO+NH_2OH+H^++e^- \rightarrow {}^*CO\text{-}NH_2+H_2O$,[4] using nudged elastic band (NEB) and climbing image nudged elastic band (CI-NEB) calculations.[29–31] Our DFT results show that the activation barrier of C-N coupling step is 0.48 eV (**Figures S63-S64**). We further validated that our transition state (TS) is true provided that the TS has only one imaginary frequency (i.e., 589.57 cm^{-1}). While surveying different reaction substrates (**Figure 2F**), we found that the FE of urea declines significantly from 63% to < 2% when nitrate (NO_3^-) is substituted with nitrite (NO_2^-). This finding was surprising as all NO_3RR reactions pathways progress through $^*NO_2$. This led us to evaluate the role of binding site competition between substrates (e.g., $^*NO_2$ vs. $^*NO_3$ vs. $^*CO_2$) and considered that the copper sites may preferentially bind nitrite over CO_2. When we reduced the nitrite concentration from 100 mM to 10 mM experimentally, we observed a 3-fold increase in urea production (**Figure S18**). To explore this phenomenon further, we calculated and compared the binding energies of $^*CO_2$, $^*CO$, $^*NO_2$, and $^*NO_3$ (**Figure S62**). The DFT calculations show that the binding of $^*NO_2$ over the single Cu atom site is much stronger than that of $^*CO_2$, with the preferential binding of nitrite at the single Cu sites would hinder CO_2 binding, limiting the CO_2RR-to-CO reaction and subsequently, urea production.

CONCLUDING REMARKS

In this study, we set out to investigate the utility of a metal organic cage material for the electrochemical co-reduction of carbon dioxide and nitrate to form C-N bonds. We found that the BIF-29(Cu) material was capable of electrochemically reducing different carbon and nitrogenous small molecules in concert to generate urea with high selectivity and activity, despite possessing single metal sites. We propose a cascade-type mechanism, where carbon dioxide and nitrate are first partially reduced at individual sites before consolidating at a single copper site to generate a $^*CONH_2$ based species that was observed experimentally through *in situ* infrared absorbance spectroscopy. Further analysis of the active site through X-ray absorption spectroscopy revealed very little structural evolution under experimentally relevant conditions, i.e., in the presence of both CO_2 and NO_3^-. Correspondingly, the atomic scale DFT calculations provided the fundamental understandings of urea formation from CO_2RR coupling with NO_3RR over the single Cu active site of Cu-BIF structure. The calculated Gibbs free energy diagram agrees well with the experiments that $^*CO$ and NH_2OH species are the potential key intermediates, generated from two separate Cu single active sites. Then NH_2OH will diffuse to the bonded $^*CO$ to produce urea via C-N coupling over one Cu single active site. This work offers a rich molecular platform to design discrete catalytic sites for carbon and nitrogen activation and further advances techniques for heteroatomic coupling in electrochemical synthesis.

ASSOCIATED CONTENT

The Supporting Information is available free of charge at https://pubs.acs.org/doi/xxxx. Experimental procedures, electrochemical characterization, and computational details (PDF).

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